Digital Logic Design

Class Activity

Name: Muneeb Lone Roll no.: 23i-2623 Sec: B

Learning Objectives

- Understand the design and operation of a 2-bit synchronous counter.
- Construct state diagrams, state tables, and excitation tables for counters.
- Derive Boolean expressions for flip-flop inputs using K-maps.
- Design and implement a circuit diagram for a 2-bit synchronous counter.

2 bit synchronous counter by JK flip flop

Step 1: Identify the number of bits and flip-flops

for step we have to identify we have

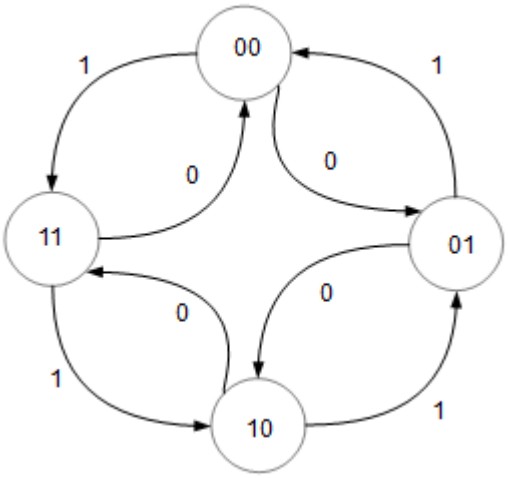
bits =__2_____

flipflop type=____JK Flip Flop____

Step 2: Write the excitation table

Qn	Qn+1	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3: Make the state diagram and state table



Current State		Next State		J1K1		JoKo	
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	1	X	1	X
1	1	0	0	X	1	X	1

Step 4: Solve the Boolean expressions

- J1 = Q0
- K1 = Q0
- J0 = 1
- K0 = 1

Step 5: Make the circuit diagram

Step 6: Simulate the circuit diagram On proteus

References:

https://youtu.be/7tDl247UK5Q?si=eivPqtP6u_hwglJv